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Unburned gas temperature measurements in a surrogate Diesel jet via two-color toluene-LIF imaging

G. Tea [a,b,*], G. Bruneaux [a], J.T. Kashdan [a], C. Schulz [b]

[a] IFP New Energy, Rueil-Malmaison, France
[b] IVG, University of Duisburg-Essen, Duisburg, Germany

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Abstract

Non-intrusive temperature measurements of the unburned fuel/air mixture in vaporized Diesel jets have been performed using two-color toluene laser-induced fluorescence (LIF). This diagnostics technique exploits the temperature-dependent spectral shift of the LIF signal which occurs after ultraviolet (UV) excitation of toluene that is added as tracer to a non-fluorescing base fuel. The method requires the determination of the ratio of LIF intensities collected by two detectors separate spectral bands. In the current study, measurements were performed in a high-pressure, high-temperature cell capable of reproducing the thermodynamic conditions in the combustion chamber of a Diesel engine during the injection event. Various aspects of the experimental set-up and the data evaluation were optimized. The temperature sensitivity of the measurement strategy is optimum at temperatures below 700 K. Temperature data acquired from two-color LIF thermometry were compared to single-color toluene-LIF measurements using an adiabatic mixing model. The latter is determined from toluene LIF-based fuel concentration measurements, the evaporation enthalpy, and thermocouple measurements of the bath-gas/ambient cell temperature prior to fuel injection. Based on simultaneous measurements with two cameras using identical optical filters a methodology to optimize the image superposition and to minimize the statistical error was developed. These measurements also allowed to determine the $1-\sigma$ precision of the two-color LIF measurement to be in the 20–40 K range.
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Keywords: Temperature measurement; Vaporizing Diesel jet; Laser-induced fluorescence

1. Introduction

Optical diagnostics have become a powerful tool to obtain detailed insight into the physical processes occurring inside the chambers of internal combustion (IC) engines, which is a prerequisite for the further development of new combustion strategies. The most important parameter for combustion in IC engines is arguably gas temperature, which has a significant influence on mixture preparation, auto-ignition, combustion, and pollutant formation. High local temperatures in the endgas can cause engine knock [1] in spark-ignition (SI) engines, and simulations have shown that even small temperature fluctuations of 10 K have a major impact on the onset of auto-ignition. Recent studies suggest that new combustion strategies such as low-temperature combustion (LTC) and

* Corresponding author at: IFP New Energy, R1060R, 1 et 4 Avenue de Bois Préau, 92852 Rueil-Malmaison Cedex, France. Fax: +33 1 47 52 66 85.
E-mail address: gabrielle.tea@ifp.fr (G. Tea).

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homogeneous-charge compression ignition (HCCI) may require control of the in-cylinder fuel and temperature distribution. Non-intrusive diagnostics are thus required to obtain in-cylinder imaging measurements with sufficient precision to evaluate the effect of spatial temperature fluctuations.

Several non-intrusive optical techniques exist to measure gas temperature such as coherent anti-stokes Raman scattering (CARS) [2] and diode-laser absorption [3–5]. The latter provide line-of-sight averaged data and typically cannot provide information on temperature inhomogeneities. Laser Rayleigh scattering has been applied for temperature imaging [6], including spark-ignition (SI) engines [7]. Rayleigh scattering is difficult to apply in practical combustion situations due to signal interference caused by strong elastic scattering from particles within the flow (e.g. soot, liquid droplets).

Temperature images can be obtained from tracer-based laser-induced fluorescence (LIF) [8]. The temperature-dependence of the collected LIF signal can be exploited to simultaneously measure temperature and fuel concentration. Tracers such as acetone [9], 3-pentanone [10], toluene [11–13], biacetyl [14], and naphthalene [15] have been used for thermometry. Recent measurements of in-cylinder unburned gas temperature were reported [16,17] with 3-pentanone two-line excitation at 277 and 308 nm. This paper reports the development and application of toluene-LIF thermometry for the vaporized part of Diesel sprays by using *n*-dodecane, a non-fluorescing fuel, which is commonly used as a surrogate Diesel fuel. Our technique is based on single-wavelength LIF excitation of toluene at 248 nm [12] and uses the temperature-dependence of the toluene-LIF emission spectrum. The main advantage of toluene over 3-pentanone for temperature measurements is that only one laser is required, reducing experimental cost and complexity. Local variations in laser fluence and tracer concentration cancel when determining the ratio of the signal of two spectral ranges of the emission spectrum.

2. Experimental

Measurements have been performed in an optically-accessible, high-pressure, high-temperature cell ($V = 1400\ \mathrm{cm}^3$) allowing the study of fuel sprays under conditions typical for Diesel engines during injection. Flow in the initially quiescent gas is induced by the spray itself. For details concerning the cell and its operation, see Ref. [18]. The injector is mounted within the top face of the cell. Five 100 mm diameter 20 mm thick sapphire windows are mounted on the remaining faces. The gas temperature was measured with a K-type thermocouple (25 μm bead) in the central region of the cell. The cell is electrically heated to 453 K to prevent water condensation.

A pre-combustion of ethylene/hydrogen/oxygen/nitrogen mixtures provides high temperature and pressure [19]. The start of injection (SOI) takes place at a specified delay after the pre-combustion when the desired temperature and pressure values are reached. For the experiments presented here, the pre-combustion mixture was chosen to completely consume oxygen providing inert conditions of the injected fuel. For all the tests reported in this paper that required pre-combustion to reach high temperature, i.e. temperature and pressure ranging from 623 K and 46.0 bar to 728 K and 53.6 bar, an ambient density of 25 $\mathrm{kg/m^3}$ was used.

A Bosch common-rail injector with a single-hole nozzle (0.2 mm diameter, 5% rounding) directed its spray downwards into the center of the chamber. Fuel (*n*-dodecane, 99.9%, doped with 10 vol.% toluene) was supplied at a pressure of 1000 bar.

2.1. Diagnostics strategy

Two-color toluene-LIF thermometry takes advantage of the temperature-dependent shape of the LIF emission spectrum after ultraviolet (UV) excitation at 248 nm. The signal ratio from two different sections of the LIF emission spectrum must be determined. For imaging, two LIF images are captured simultaneously with two ICCD cameras that are equipped with appropriate (different) filters. A calibration enables to determine temperature from the LIF signal ratio. The calibration is based on toluene-LIF emission spectra that are acquired under well-controlled conditions at variable temperature [12]. In our case, spectrally-resolved in-cylinder measurements were used where the temperature was determined from a thermodynamic analysis of the pressure trace. When applying the calibration, the spectral response of the entire detection system, i.e. the transmission characteristics of the optical components (e.g. cell windows, optical filters, lenses, photo cathodes) must be taken into account. Further details of the calibration method used are provided in Refs. [20,21]. For the cell windows, the transmission curves were measured using a spectrometer coupled to an optical fiber using a deuterium/halogen light source. Manufacturer-supplied data was used to characterize all other optical components.

Two-color toluene-LIF thermometry has been successfully used in atmospheric-pressure flows [20,22], theoretically assessed [23] and applied [21] to SI engine conditions. The strong reduction in total signal with increasing temperature results in weak signals at temperatures above 700 K and thus requires detection with a high dynamic range in order to perform measurements over a wide temperature range without changing the experi-

mental settings. This is a potentially limiting factor when applying this technique to Diesel jets that are characterized by high temperature gradients. However, with careful consideration of various aspects of the measurement technique, it is possible to optimize the toluene-LIF signal to enable measurements even in such challenging environments.

Preliminary experiments revealed that a toluene concentration of 10% was required to provide sufficient signal. Despite this high concentration, the measurements did not suffer from laser attenuation due to the small amount of fuel injected into the cell. The spectroscopic purity of the dodecane fuel was investigated since impurities cause 'background' fluorescence. The background level was below 10% compared to the toluene-LIF signal for all measurement conditions and thus considered negligible.

Toluene-LIF thermometry relies on photophysical processes occurring in the linear fluorescence regime. Because saturation at low temperatures is reached at lower fluences [22], linearity checks were performed at 453 K which is the lowest temperature in our experiments. To provide a homogeneous toluene concentration inside the cell the fuel–tracer mixture was injected 20 s before the laser measurement allowing sufficiently long mixing time. Tests with excitation laser fluencies in the 20–100 mJ/cm^2 range were performed. In order to maximize the signal-to-noise ratio we chose 50 mJ/cm^2 as the highest possible laser pulse fluence that still maintained linear fluorescence response.

Figure 1 shows the optics set-up relative to the high-pressure cell. A KrF–excimer laser (Lambda Physik, CompexPro 102) was used to excite toluene at 248 nm. A vertical laser light sheet (47×0.4 mm^2) entered the high-pressure cell. Dark coated metal masks with holes in the shape of the laser beam that were mounted on the inner side of the laser entry and exit windows helped to reduce the detection of elastically scattered laser light. The signal was collected through two opposite windows with two intensified CCD cameras (Princeton PI-MAX, 1024×1024 pixel) equipped with filters with high transmission ($T > 70\%$) and fast achromatic UV lenses ($f = 45$ mm, $f_{\#} = 1.8$, EADS Sodern). Two different experimental configurations were used for our experiments that differ in the choice of optical filters used for LIF detection. The first set-up (denoted hereafter as config.1) was used for temperature measurements and enabled the detection of toluene fluorescence in two different spectral intervals. One filter (280 nm, full-width at half maximum, FWHM = 20 nm Semrock) was centered at the peak of the toluene-LIF emission spectrum, the other filter (HP300 nm, Semrock) was used for the detection of signal in the long-wavelength tail of the toluene-LIF spectrum. For config.2, identical filters (280 nm band pass filters) were used for both cameras to investigate various aspects of the measurement technique as will be discussed later in the paper.

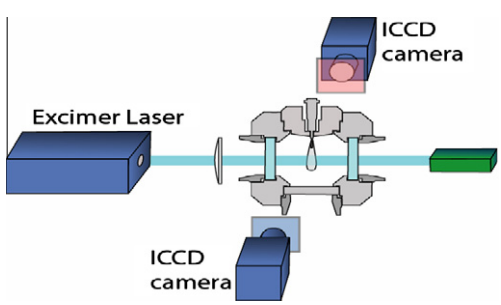

Fig. 1. Experimental set-up.

2.2. Experimental methodology

Prior to each series of temperature measurements with config.1, config.2 was used for a series of standardization experiments. Therefore, the cell was homogeneously filled with the vaporized fuel/tracer mixture. LIF images of the laser sheet profiles were then acquired with both cameras. The comparison of the pair of images allowed to correct for variations in pixel-to-pixel sensitivity of both cameras, as discussed later. After acquiring images of homogeneous toluene LIF distributions, LIF images in the Diesel jet (624 K and 46 bar ambient condition) were measured after pre-combustion. As discussed later, the pair of single-shot LIF images detected in the identical spectral band served as reference images to perform (1) spatial adjustments of the entire series of images and (2) to calibrate differences in the overall sensitivity of the two CCD cameras.

Filters were then used according to config.1 for two-color temperature measurements. Background images were acquired under the same cell conditions but without injection. In addition, dark current background was subtracted. The image pairs obtained with fuel injection were then

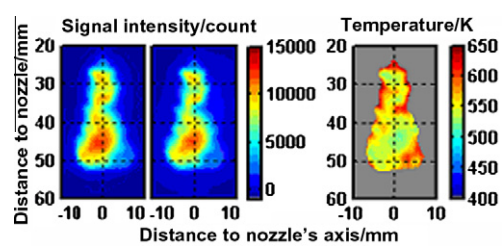

Fig. 2. Toluene-LIF images obtained with the 300 nm high pass filter (left) and the 280 nm band pass filter (center). Resulting temperature distribution at an ambient temperature of 600 K (right).

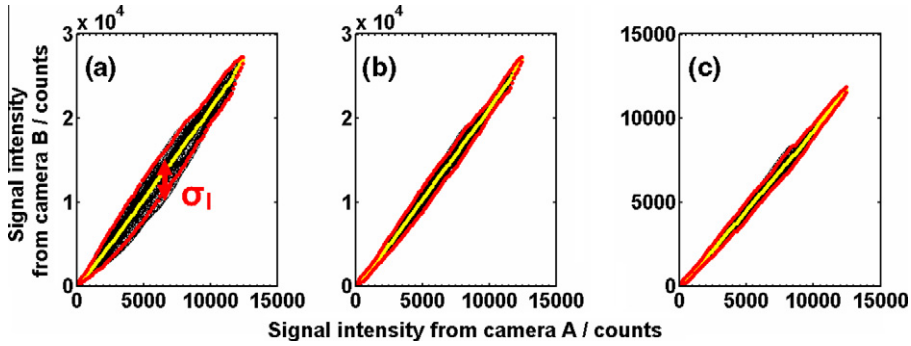

Fig. 3. Signal intensity correlation for an image pair obtained in config.2. (a) After spatial adjustment from reference grid image; (b) after optimization of the spatial adjustment; (c) after optimization of the spatial adjustment and flat field correction.

corrected for variations in pixel sensitivity and for spatial overlap using the data obtained from config.2. Additionally, the images were filtered for noise reduction as described in more detail in Section 3. Finally, the LIF-ratio image was determined which was then converted into a temperature image using the calibration data. An example of toluene-LIF images and the resulting temperature distribution is presented in Fig. 2.

3. Data post processing

Various aspects of the data processing were investigated using the LIF images obtained with config.2. Theoretically, both images should be identical besides a factor that would be attributed to different detector sensitivities or optical efficiencies. A pixel intensity correlation plot should therefore result in a straight narrow line. Figure 3, however, shows significant dispersion around the mean which is due to imperfect superposition of the two images, variations in pixel-to-pixel sensitivity for both cameras, and shot noise. Additionally, the two detection paths (including windows, filters, and lenses) might have slight spatial variations in transmission properties. The image pair can thus be used to optimize image superposition, standardize the sensitivity of both detection systems, and evaluate the sources of various measurement uncertainties and their effects on the precision of the technique.

3.1. Image superposition

Calculating the LIF-intensity ratio relies on an accurate superposition of the two images. In the present work, the superposition was initially achieved by using a reference grid imaged by both cameras. A comparison of the two images yields a transformation matrix that is then applied to adjust one image relative to the other in terms of translation, rotation and scaling. Figure 3a reveals significant dispersion in the scatter plot (σ_I) even though the standard transformation matrix has been applied. Discretization of the scatter plot on the x-axis (using an interval width of 100 counts) enabled the determination of the mean (yellow[1] dots) and $\pm 2\sigma$ standard deviation (red dots) as shown in Fig. 3. The transformation matrix was further optimized by using the magnitude of the dispersion (σ_I) as optimization criterion leading to a dispersion (σ_I) that was reduced by 70% (Fig. 3b).

3.2. Flat field correction

The differences in the pixel sensitivities of both cameras also contribute to the dispersion. The correction procedure was based on toluene-LIF images in homogeneous conditions inside the cell, using config.2. Because both images should be identical, it was possible to determine differences in sensitivity on a pixel-by-pixel basis from averaged images (20 individual images to reduce shot noise). The LIF-ratio image was then determined and used as a correction image. The correction was performed by multiplying one of the Diesel jet LIF images by this correction image (reference ratio image). The additional effect of this correction can be seen by comparing Fig. 3b and c. The dispersion (σ_I) was again reduced by 50% when applying the reference ratio image, showing that although shot noise tends to reduce the efficiency of this correction, the effect on measurement precision is beneficial.

3.3. Adjustment of the camera sensitivities

The pair of Diesel jet LIF images obtained with config.2 was also used to correct for differences in sensitivity between the two detectors.

[1] For interpretation of color in Fig. 3, the reader is referred to the web version of this article.

The mean values, represented by yellow dots in Fig. 3c, reveals the difference in sensitivity of one detector relative to the other (which deviates slightly from a constant factor as a function of intensity). The resulting function was thus used to correct the LIF ratios when processing the images from config.1. With this methodology the fluorescence ratio becomes independent of sensitivity differences of the two cameras. The temperature-versus-LIF-intensity-ratio calibration curve derived from toluene emission spectra and corrected by the transmission characteristics of the collection optics (determined from data reported in Ref. [12]) can therefore be *directly* applied to the LIF-ratio images and does not require an additional calibration measurement at conditions where the temperature is known. The calibration method is described in Refs. [20,21]. With the two detectors matched in sensitivity, the technique does not require in situ calibration for the determination of temperature.

3.4. Error analysis – determination of random error

The error sources that affect the precision (random error) of the measurement include errors due to imperfect superposition of the two images, differences in pixel-to-pixel sensitivity of the two cameras and camera shot noise. Because the comparison of the two LIF images in the jet from config.2 is also affected by these error sources, it can be used to determine the measurement precision as follows.

A pair of toluene-LIF images of the Diesel jet is acquired using config.2 which was then used to determine a LIF-ratio image. For all pixels, the LIF ratio should be unity as long as significant signal is measured. This is, however, not the case and there is some dispersion of the ratio. Its magnitude indicates the contribution of the different error sources discussed before. This analysis of the ratio image is similar to the image-pair analysis displayed in Fig. 3, except that the magnitude of the dispersion is expressed in terms of the ratio. This is more convenient because the measured temperature is directly related to the LIF ratio by using the toluene-LIF calibration data.

In the present experimental set-up, the following expression (Eq. (1)) corresponds to the fit relating temperature, T, and the LIF ratio, R applying the calibration method in Refs. [20,21]:

$$T = 238\ \mathrm{K} \times \ln(R) + 626\ \mathrm{K} \tag{1}$$

As a consequence, the dispersion of the LIF ratio can be expressed in terms of a dispersion in temperature (Eq. (2)):

$$dT = 238\ \mathrm{K} \times (dR/R) \tag{2}$$

The measurement precision (statistical error) was then determined from the ratio image obtained in config.2 using Eq. (3):

$$\sigma_T = 238\ \mathrm{K} \times (\sigma_R/R) \tag{3}$$

where σ_T is the temperature measurement precision and σ_R the standard deviation of the values of R in the ratio image.

An example of the evolution of σ_T versus the signal intensity of one of the image pairs obtained in config.2 is presented in Fig. 4. The values presented in this figure were obtained after applying the various correction procedures described above. Additionally, both images were numerical filtered to reduce the effects of shot noise and two types of filter were investigated, a 'median filter' and a 'mean filter'. The filter box size was $2 \times 2\ \mathrm{mm}^2$ as discussed in more detail in the following. The results show that the statistical error is maximal for low levels of LIF signal, around 23 K with the 'mean filter', and decreases to below 10 K in regions of the jet where the LIF-signal intensity is maximal. This is consistent with the relative contribution of shot noise in the regions of low or high signal intensity. These results also show the difficulty of applying the two-color toluene-LIF thermometry for measurements in a Diesel jet configuration. In the jet periphery, the measurement precision is low as a result of low fuel concentration and high temperature which leads to a decrease in LIF signal strength.

After optimizing the image superposition and applying the flat field correction, the main contributor to statistical error is shot noise. In order to reduce its contribution, the LIF images were numerically filtered. In the present work, the effects of the type of the numerical filter and its filter box size were investigated. Figure 5 shows the

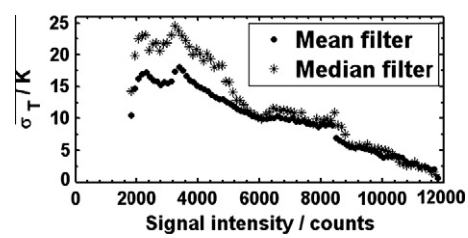

Fig. 4. Statistical error σ_T as a function of signal intensity for an image pair obtained in config.2.

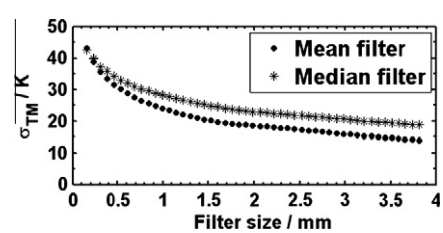

Fig. 5. Evolution of the maximal measurement statistical error as a function of the filter box size, for two types of numerical filters, calculated with an image pair obtained in config.2.

evolution of the maximum measurement statistical error σ_T^M as a function of the filter box size for two types of filters. The value of σ_T^M presented in this figure corresponds to the maximum value obtained in the LIF-ratio image, corresponding to regions of low signal intensity in the images as shown in Fig. 4 (in Fig. 4, σ_T^M is approximately 23 K with the mean filter). The filter box size used for filtering is expressed in mm^2 (rather than pixels, 1 pixel = 0.08mm). Figure 5 shows that the statistical error σ_T^M decreases with increasing filter box size and is lower with the 'averaging filter'. It shows that there is a compromise between measurement precision and spatial resolution. In the present work a filter box size of $2 \times 2\ mm^2$ was selected.

3.5. Error analysis – systematic error evaluation

In order to evaluate the accuracy of the temperature measurement (i.e. the systematic error), it was necessary to compare the temperatures measured by the two-color toluene LIF technique against a known (i.e. reference) temperature. The bath-gas temperature T_{TC} (measured by thermocouple) in the central zone of the cell prior to fuel injection was used as a reference. When measurements were carried out in fully homogeneous mixtures within the electrically-heated cell at 453 K, the homogeneous toluene-LIF temperature data can be directly compared to T_{TC} and thus allows to determine the measurement accuracy at this low temperature.

Reference measurements at higher temperature, however, could not be realized in the same way. Therefore, a different strategy was developed that is based on fuel injection into the hot bath gas that was heated by the pre-combustion. The temperature was again known from thermocouple measurements in the absence of a spray. In the presence of a spray, the temperature is reduced locally due to evaporative cooling. Assuming adiabatic mixing between ambient gas and fuel, the deviation from T_{TC} scales with the local fuel concentration and can be determined in an iterative way from a single-line measurement of toluene LIF. A relatively long delay between the end of injection and image acquisition ensured sufficient time for complete evaporation to circumvent interference from liquid fuel. In a zero-order approximation, LIF intensities obtained with the filter (HP300) were interpreted as maps of fuel number densities assuming spatially invariant fluorescence quantum yields according to Ref. [24] which is based on a comparison of local intensities to those measured after injection and homogenization of a known fuel mass. From the local concentration the local temperature reduction can be calculated from the fuel mass-related evaporative cooling via the adiabatic mixing model [25]. The resulting zero-order temperature field was used to replace the initial assumption of constant fluorescence quantum yields by temperature-dependent values leading to a first-order correction in local fuel mass. This again led to improved temperatures and this procedure was carried on in an iterative way until no further changes occurred. The result was a temperature map from single-line LIF assuming adiabatic-mixing that will be denoted hereafter as T_{AM}.

There are various sources of uncertainty in this procedure. Firstly, uncertainties arise from the determination of the injected fuel mass and the effect of temperature on the fluorescence yields. The authors believe, however, that the main source of uncertainty is due to the adiabatic mixing model itself. The model is based on the hypothesis that mixing is adiabatic, which in the present experiment is a reasonable assumption since neither significant wall heat losses nor radiative heat losses are expected. However, errors are likely due to an incorrect determination of the boundary conditions: the ambient temperature and the temperature of the pure vaporized fuel. The uncertainty of the ambient temperature is given by the accuracy of the thermocouple measurement. However, determination of the pure vaporized fuel temperature is not straightforward. A value of 473 K was selected according to the thermodynamic model presented in Ref. [12]. Consequently, the error of this temperature determination is believed to increase with increasing deviation from ambient temperature T_{TC} and, thus, fuel concentration.

A comparison is made between the two-line LIF temperatures (T_{LIF}) and the temperatures derived from the single-line LIF using an adiabatic mixing model (T_{AM}), within the high-temperature regions of the jet that are located close to the jet periphery. Around these regions the temperature is close to the ambient/bath-gas temperature (T_{TC}) which is therefore expected to be reasonably accurate. The results of this comparison derived from six individual measurements are shown in Fig. 6. It reveals that below 700 K, measurement accuracy (systematic error), determined from the difference between T_{LIF} and T_{AM} at the edge of the jet is better than ± 30 K. A more accurate determination of the systematic error would require a higher number of samples for statistical analysis. For temperatures higher than 700 K, the difference between the two temperatures increases significantly. This deterioration of the measurement accuracy at high temperature is due to a significant decrease of the toluene-LIF signal, resulting in a reduced signal-to-noise ratio from 10 to 5 in the jet periphery at ambient temperatures of 624 and 728 K, respectively. Higher signal-to-noise levels would be required to enable temperature measurements at higher temperatures.

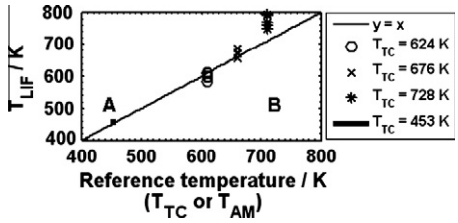

Fig. 6. Comparison between results of two-color LIF thermometry (T_{LIF}) with (A) thermocouple (T_{TC}) measurements in the electrically-heated cell, and (B) single-line LIF measurements with the adiabatic mixing model (T_{AM}) applied to the jet periphery where the fuel concentration equals 0.4 kg/m^3. Measurements with different bath-gas temperatures: $T_{TC} = 624$, 676, 728 K). Injection duration: 600 μs, detection 1.5 ms after start of injection.

4. Discussion

Diesel jet temperatures corresponding to one injection event (i.e. single-shot LIF images) obtained by LIF imaging (T_{LIF}) are compared to T_{AM} in Fig. 7. Unlike Fig. 6 where the temperature was calculated within the periphery of the jet at a given fuel concentration, the data presented in Fig. 7 corresponds to temperature measurements which have been made within the entire jet with variable fuel concentrations. For a given range of T_{AM} values corresponding to specific locations in the image, mean averaged and standard deviations of T_{LIF} values in the corresponding locations in the image are presented. The results reveal satisfactory agreement when comparing the average temperature measured by LIF imaging with the corresponding T_{AM} value although higher discrepancies are observed at lower temperatures. These results show that the temperatures determined via single-line LIF imaging with the adiabatic mixing model are comparable to the results

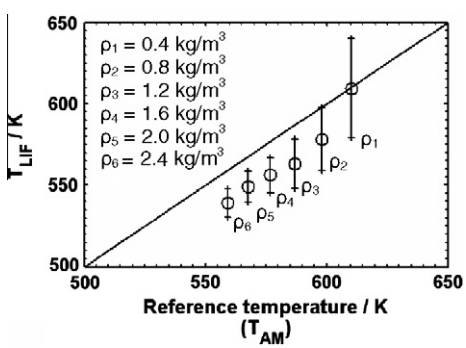

Fig. 7. Comparison of LIF ratio temperature (T_{LIF}) and T_{AM} in the whole jet at different fuel concentrations (ρ) for one injection event with an ambient temperature of 624 K (T_{TC}). The circles correspond to average T_{LIF} values over a range of T_{AM}, while the crosses correspond to the standard deviation.

of the two-color LIF thermometry. The discrepancy observed at lower temperatures is attributed to the fact that the adiabatic mixing model yields increasing errors with increasing fuel concentration and, thus, decreasing temperature. The temperature standard deviations determined from the two-color LIF thermometry (T_{LIF}) compared to T_{AM} are consistent with the statistical error evaluation in Fig. 4. In particular, in the lower temperature regions of the jet the standard deviation is below 10 K while it increases to 30 K in the higher temperature zones. These values compare well with estimates of the statistical error presented earlier (Fig. 4) and thus indicate that the major source of error when comparing the two methods is most likely due to this statistical error.

As a result, it seems reasonable to conclude that in the configuration studied here, local temperature fluctuations in the jet are most likely due to the presence of fuel concentration fluctuations arising from temperature inhomogeneities which are related to the mixing process. Overall, the initial results are promising and indicate that the two-color toluene-LIF thermometry is a highly valuable tool for temperature imaging and, thus, an aid for gaining an improved understanding of the mixing and auto-ignition processes in Diesel jets.

5. Conclusions

Measurements of unburned gas temperature in an evaporating Diesel spray have been performed using two-color toluene-LIF temperature imaging. Various aspects of the technique were studied and a detailed uncertainty analysis has been performed. Using single-wavelength excitation at 248 nm and two-color detection, the measurement precision (σ) of the technique is in the 20–40 K range.

Various sources of random and systematic errors were examined and methods are reported which help to significantly improve measurement precision and accuracy. In particular, improved data processing algorithms have been developed including for example, an iterative optimization procedure which allows precise superposition of the LIF image pair. Flat field image corrections are also performed to compensate for variations in pixel-to-pixel sensitivity of the two detectors. Finally, the calibration procedure also takes into account the spectral characteristics of the various optical components (i.e. windows, optical filters, lenses and spectral response of the image intensifiers).

The measurement accuracy was evaluated by comparing the temperature measured by the two-color toluene LIF technique with a known ambient/bath-gas temperature measurement. A straightforward comparison was possible for one temperature (453 K) within the electrically-heated

cell under homogeneous conditions without pre-combustion. At higher temperatures an indirect comparison was performed between the jet temperatures derived from the two-color LIF ratio with those derived from a single-color LIF measurement using an adiabatic mixing model.

The results reveal that toluene two-color LIF thermometry has limitations at temperatures above 700 K. It might be possible to perform measurements at higher temperatures by using higher laser fluence in a partially-saturated situation to improve LIF signal-to-noise ratio, thus giving up some of the spectroscopic rigidity. Furthermore, the results presented suggest that local temperature fluctuations in the Diesel jet are most likely due to the presence of fuel concentration fluctuations arising from temperature inhomogeneities which are related to the mixing process.

In the present work, experiments were performed in conditions close to adiabatic mixing conditions which enabled the validation of the technique. An application of the technique appears to be promising for more complex, i.e. non-adiabatic conditions as well.

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